Triage Report — Sentinel Intelligence

1. Submission Summary

Sentinel Intelligence proposes an AI Foundry-based Operations Assistant for GMP manufacturing that detects anomaly/deviation events from SCADA/MES/IoT signals, enriches them with batch/equipment context, grounds decisions in SOP/BPR/CAPA history via RAG, generates CAPA recommendations and audit-ready reports, and keeps a human approval step before work-order execution.

- **Industry:** LS / Supply Chain
- **Use Case: Predictive maintenance Approach — Operations Assistant** for **Deviation Management & CAPA in GMP Manufacturing**
- **Track:** Not declared; closest fit is **A** (GitHub + Azure + optional Microsoft Fabric)
- **Stage:** Submission Review

Cross-cutting checks

Check	Status	Notes
Environment Track declared	**Fail**	No A / B / C / D track is explicitly stated. The stack shown (Azure Functions, Azure AI Search, Foundry Agent Service) is most consistent with **Track A**, but GitHub and track declaration are missing.
Industry Use Case aligned	**Pass**	Strong match to **LS / Supply Chain** Section 3.10, specifically **Predictive maintenance Approach — Operations Assistant**. The submission extends the reference use case into deviation handling and CAPA, which remains aligned to the stated GMP problem.
Azure WAF aligned	**Warn**	Good start on architecture layering and integration, but explicit coverage of **Reliability, Security, Cost Optimisation, Operational Excellence, and Performance Efficiency** is incomplete. Security, resiliency, and cost controls are not sufficiently defined.
Azure WAF AI Pillar aligned	**Warn**	Strong on **Agent Design** and **Grounding/RAG**; partial on **Model Lifecycle** via “evaluation” and “governed deployment.” Missing explicit **Responsible AI**, **AI Observability**, prompt-injection defenses, and confidence/evidence thresholds.
Microsoft Security & Monitoring	**Warn**	Audit logging is included, which is good for traceability, but the submission does not show identity architecture, RBAC, secrets handling, SIEM/monitoring, alerting, or Azure security controls across the stack.

2. Criteria Scoring

Architecture

Dimension	Score	Justification
Clarity & Flow	**8/10**	The detect → context enrichment → compliance agent → CAPA/audit agent → human approval → work order/audit record flow is clear and easy to follow. Agent roles and data sources are visible, though exception paths and runtime decision logic are still high-level.

Platform Fit	**7/10**	Azure Functions, Azure AI Search, and Foundry Agent Service are appropriate for event ingestion, RAG, and multi-agent orchestration in a GMP scenario. However, the required hackathon **track is not declared**, and the architecture does not show the full track-aligned developer/platform setup.
Data / Governance / Security	**6/10**	Data sources and governed retrieval are well identified: SCADA, MES, BPR, SOP, CMMS, and CAPA history. Human approval and audit logging support governance, but identity, access control, encryption, retention, classification, private connectivity, and content safety are not described.
Reliability / Performance / Cost	**5/10**	The business target of reducing decision time to under 5 minutes shows performance intent, but the design does not define retries, queues, failure recovery, fallback behavior, model timeout handling, cost controls, or token/resource optimization.
Scalability / Integration / Provisioning	**7/10**	Enterprise integrations are clearly identified and the major Azure services are named, which supports scalability and provisioning discussion. Still, there is no IaC/deployment approach, API contract detail, environment topology, or repeatable provisioning design.

Architecture Subtotal: 33/50

Use Case

Dimension	Score	Justification
Value & KPI Impact	**9/10**	This is a high-value regulated process with clear business pain: slow deviation analysis, inconsistent decisions, audit burden, and production loss. KPI impact is explicit (decision time, QA effort, handling errors, inspection readiness), though value quantification could be more evidence-based.
Innovation	**8/10**	The combination of predictive signal interpretation, GMP compliance validation, CAPA recommendation, and audit-trail generation is more differentiated than a basic chatbot or anomaly detector. It shows a thoughtful multi-agent design tied to a real regulated workflow.
AI Fit	**8/10**	AI is well suited for contextual retrieval, historical pattern interpretation, recommendation drafting, and evidence packaging. The mandatory human approval step is especially important for GxP, but responsible AI controls are not explicit enough yet.
UX Simplicity	**6/10**	The target process simplifies operator work by delivering a structured decision package and prefilled actions. However, the actual user interface/channel is not described, so ease of use, explainability, and approval ergonomics are hard to assess.
Build–Scale–Reuse	**7/10**	As an MVP, this is feasible if narrowed to one asset type, one deviation class, and a small set of SOP/CAPA sources. The retrieval and agent patterns appear reusable, but reusable MCP assets/connectors and production-scale constraints are not yet concretely defined.

Use Case Subtotal: 38/50

Total

Total Score: 71/100

3. Strengths

- Strong alignment to the defined **LS / Supply Chain — Predictive maintenance Approach** with a clear GMP problem statement and explicit stakeholder set (**Value & KPI Impact**, **Industry Use Case aligned**).

- The end-to-end architecture is easy to understand and shows a sensible sequence from detection through compliance reasoning to human approval and action execution (**Clarity & Flow**).
- Good use of Microsoft capabilities for the problem: **Azure Functions** for event-driven ingestion, **Azure AI Search** for RAG, and **Foundry Agent Service** for orchestration (**Platform Fit**).
- Governance-aware design choices such as grounding on validated SOP/BPR/CAPA history, mandatory human review, and audit logging are well matched to regulated manufacturing needs (**Data / Governance / Security**, **AI Fit**).
- Integration with MES, SCADA, CMMS, QMS, and asset history makes the concept operationally relevant rather than an isolated demo (**Scalability / Integration / Provisioning**).
- KPI intent is clear and business-facing, especially around reducing decision latency, manual QA effort, and inspection-preparation effort (**Value & KPI Impact**).

4. Gaps & Risks

- The submission does not explicitly declare the hackathon **environment track**, which is a direct compliance gap against the evaluation checks (**Environment Track declared**, **Platform Fit**).
- Security architecture is under-specified: no clear Entra ID/managed identities, RBAC model, network isolation, encryption, secrets handling, or document access controls are shown (**Data / Governance / Security**, **Microsoft Security & Monitoring**).
- Reliability design is too thin for a GMP-critical workflow: there is no visible retry strategy, queueing, dead-letter handling, fallback mode, or degraded/manual-only operating mode (**Reliability / Performance / Cost**, **Azure WAF aligned**).
- Cost and performance efficiency are not addressed for continuous sensor/event workloads and agentic RAG calls; no caching, model routing, latency budgets, or token controls are shown (**Reliability / Performance / Cost**).
- Responsible AI coverage is only partial: the deck lacks confidence thresholds, evidence gating, hallucination controls, prompt-injection defenses, agent observability, and model/version lifecycle details (**AI Fit**, **Azure WAF AI Pillar aligned**).
- The user experience is process-level rather than interface-level; without a concrete operator surface, the simplicity and trustworthiness of approvals cannot be fully judged (**UX Simplicity**).
- Provisioning readiness is incomplete because the architecture lists services but not repeatable deployment mechanisms such as IaC, environments, dependencies, or monitoring setup (**Scalability / Integration / Provisioning**).

5. Triage Verdict

Good — 71/100

6. Top Actionable Improvements

- **Declare the environment track explicitly and align the architecture to it**—most likely **Track A**—including where GitHub, CI/CD, and deployment/evaluation workflows fit in the solution lifecycle (**Platform Fit**, **Environment Track declared**).
- **Add a concrete security and governance layer**: Entra ID/managed identities, role-based access for operators/QA/compliance, Key Vault, private endpoints/VNet, encryption, data classification/retention, and controlled access to SOP/BPR/CAPA content (**Data / Governance / Security**, **Microsoft Security & Monitoring**).
- **Define production-grade resiliency and cost controls** such as event queueing, retries, dead-letter handling, model fallback, latency SLOs, caching, token budgets, and expected per-event cost/performance tradeoffs (**Reliability / Performance / Cost**, **Azure WAF aligned**).
- **Strengthen the AI assurance story** with evaluation metrics, evidence-backed recommendation thresholds, prompt-injection/content-safety controls, output observability, versioning/rollback, and explicit escalation for low-confidence cases (**AI Fit**, **Azure WAF AI Pillar aligned**).
- **Make the MVP and UX more concrete** by showing one target asset/equipment scenario, one operator approval interface, and a sample decision package with rationale/evidence mapped to the claimed KPIs (**UX Simplicity**, **Build–Scale–Reuse**, **Value & KPI Impact**).

Important Notice — AI-Assisted Evaluation

This report has been produced using an AI-powered triage system developed by Capgemini as part of the Microsoft Agentic Industry Hackathon 2026 evaluation process. While every effort has been made to ensure consistency and quality, the scores and assessments contained herein are indicative only and should not be treated as definitive or final judgements. AI-generated evaluations may contain inaccuracies, omissions, or contextual misinterpretations.

This report is intended as a supporting tool to assist Capgemini and Microsoft reviewers and also participating teams, judges, and Partner Solution Architects (PSAs) in identifying teams that are currently active and assessing relative submission strength, with the aim of informing — not replacing — human decision-making on stage progression. All final advancement decisions will be made by qualified human reviewers from Capgemini and Microsoft, who will use this report as one input alongside their own professional assessment.

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